GovScheme AI

AI-Powered Government Scheme Assistance Platform · Find · Check · Prepare · Apply

Final Year Major Project
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PROBLEM

India runs 4,000+ welfare schemes, yet most citizens miss benefits — they can't tell which schemes they qualify for, which documents are needed, or how to apply.

SOLUTION

A single web platform that **matches** a user's profile to schemes with explainable rules, **verifies document readiness** against their personal vault, and **guides** them to apply with AI assistance.

4,703	24,485	6	13+	100%
REAL SCHEMES (MYSCHEME)	SCHEME–DOCUMENT REQUIREMENTS	BACKEND MODULES	CORE FEATURES	EXPLAINABLE ELIGIBILITY

Technology Stack

Frontend: React 18 + Vite · **Backend:** Django 5 + DRF (REST APIs, JWT auth) · **Database:** SQLite (4,703 seeded schemes) · **AI:** Gemini API (chatbot + OCR) · **PDF:** ReportLab · **Email:** Gmail SMTP · **Data:** scraped myScheme portal

Core Features

1. **Scheme Discovery** — 4,703 real schemes with search, category/state filters, save & side-by-side compare.
2. **Eligibility Engine** — deterministic rule evaluator (income, age, state, caste…): per-rule PASS/FAIL reasons + 0–100% match score.
3. **Personalized Dashboard** — eligible & recommended schemes ranked by profile match, with eligibility badges.
4. **What-If Simulator** — test hypothetical changes ("income rises to ₹8L") and instantly see schemes gained or lost.
5. **Profile Completion Meter** — completeness % with live scheme-impact per field ("Add State → +1,561 schemes") and jump-to-section chips.
6. **Document Vault** — per-user secure storage for Aadhaar, PAN, income proof, caste certificate, land records, etc.
7. **OCR Scanner** — Gemini Vision reads an uploaded document photo, classifies its type and extracts key fields automatically.
8. **Document Readiness** — maps every scheme's required docs to the vault: **In Vault / Missing / Manual** states + "Application Ready" indicator.
9. **Application Certificate** — downloadable PDF with unique verification ID; public verify page validates authenticity.
10. **Email Notifications** — new schemes are matched against all profiles; eligible users are auto-emailed, with a delivery log.
11. **Admin Panel** — staff-only portal: scheme + eligibility-rules builder, required-docs builder, notification log.
12. **AI Chatbot** — profile-aware answers grounded in retrieved scheme context, with offline template fallback.

Live Demo Flow

Login (JWT) → Dashboard: badge-ranked schemes + completion meter → Scheme detail: rule-by-rule eligibility breakdown → What-If Simulator → Document Readiness: In Vault / Missing / Manual → Vault: OCR scan & auto-classify → Certificate → public verification → Admin: create scheme → auto-email eligible users → AI Chatbot Q&A

What Makes It Stand Out

Explainable by design — eligibility is legal criteria, not probability: every rule shows PASS/FAIL with a human-readable reason.

Readiness, not just uploads — each scheme's free-text document list is mapped to the vault (multi-type matching: "Proof of Identity" = Aadhaar OR PAN), with "Manual" flags for non-digitizable items.

Complete loop — admin publishes a scheme → engine auto-detects eligible users → emails them → they download a verifiable certificate (public verification page).

Grounded AI — the chatbot answers cite real retrieved schemes and the user's profile; OCR feeds the vault. Both fall back gracefully if the AI API is unavailable.

Run: `cd backend → python manage.py runserver 127.0.0.1:8001` · `cd frontend → npm run dev` → `localhost:5173`
Demo login: `adwait / adwait@123` · **Admin:** staff account for Admin Panel access